

Curriculum Intelligence Engine — Week 3: Core NLP Model (redo)

Group 9 · Roll Nos: J048, J049, J050, J052, J053 · AI University NLP Course Project 2026 · redone 9 Oct 2026

CODE

```
Repository : https://github.com/jainamsethia/Curriculum-Intelligence-Engine
Tag        : week-3-redo
Commit     : 2d41ffa154bd37c4217c20468f7c8d07d4f7463b
Run        : python -m curriculum_engine.run week3      (after: run data, run week2; NMIMS_ZIP=<path to "B TECH.zip">)
Core code  : curriculum_engine/overlap.py, redundancy.py, prereq.py, diff.py, topics.py, emerging.py, engine.py
Folders    :
.gitignore
README.md
curriculum_engine/  the NLP engine (parser, data, overlap, redundancy, prerequisites, diff, Bloom, evaluation)
labels/             LLM-annotated reference labels (ids + labels per pass + consensus)
legacy/             old Week 3-4 scripts and reports (history only)
legacy_mit/         old MIT OCW scripts (not used)
reports/            report templates (numbers are placeholders)
requirements.txt
results/            metrics JSON + figures per week
submissions/        per-week submission folders
tests/              unit tests (pytest)
tools/              report builder, report checker, submission packer
```

Feedback addressed

Instructor comment	What we changed	Evidence
Week 3: "Progress relative to previous week is minimal"	Every capability of Week 2 gets a semantic core model, evaluated on the same split, labels and synthetic cases; the progress table below compares them metric by metric.	results/week3/summary.json
Week 4: "overlapping content, missing pre-requisites, detect redundancies"	Overlap, redundancy and missing prerequisites are the three core models of this week, with flag precision on reference labels.	curriculum_engine/overlap.py, redundancy.py, prereq.py
Week 2: "value to NMIMS"	Example output below is on real NMIMS courses; prerequisite checks follow each programme's semester structure.	results/week3/examples.json

Progress vs previous week (same test split, same metrics)

Capability	Metric (test split)	Week 2 baseline	Week 3 core
1 Overlap	Spearman [95% CI]	0.708 [0.62, 0.78]	0.667 [0.55, 0.76]
	ROC-AUC any overlap [95% CI]	0.959 [0.93, 0.98]	0.932 [0.88, 0.97]
	F1 at validation threshold	0.774	0.746
2 Redundancy	injected near-duplicates found in top 20	0.600	1.000
3 Prerequisites	removal recall [95% CI]	1.000 [1.00, 1.00]	0.880 [0.80, 0.95]
	false alarms, intact programme	0.470 [0.34, 0.59]	0.036 [0.00, 0.08]
	false alarms after renaming the course	1.000 [1.00, 1.00]	0.120 [0.06, 0.20]
4 Curriculum diff	precision / recall of changes	0.990 / 1.000	0.998 / 0.956
	precision / recall with renamed courses	0.793 / 1.000	0.993 / 0.956
5 Bloom level	accuracy [95% CI]	0.776 [0.72, 0.83]	0.776 [0.72, 0.83]
	within one level / macro-F1	0.871 / 0.756	0.871 / 0.756
2 Redundancy	precision of flags (reference labels, sample of 20)	0.250	0.800
3 Prerequisites	precision of flags (reference labels, sample of 20)	0.050	0.100
6 Emerging topics	precision of flagged gaps	-	1.000
7 Topic labels	share of labels accepted	-	0.700

1. Core models

- **Overlap (capability 1).** Each course is split into chunks (its units; outcomes when it has no units) and embedded with all-MiniLM-L6-v2. Every chunk of A is matched to its most similar chunk of B and vice versa (MaxSim). The soft score is the mean best-match similarity; matched units (cosine ≥ 0.550 , chosen on validation) are the common topics, unmatched units are unique to each course, and the share of matched units gives the overlap percentage.
- **Redundancy (capability 2).** Inside one programme, every pair of different courses is reduced to its most similar unit pair; pairs above 0.548 (tuned on validation) are flagged with the two unit titles as evidence.
- **Missing prerequisites (capability 3).** Each prerequisite phrase is resolved by embedding similarity to course names, preferring courses taught earlier in the same programme (threshold 0.750). If no course matches, the phrase is checked as a topic against every unit of earlier-semester courses (threshold 0.400). Flags: "not offered earlier" and "assumed topic not taught". Both thresholds were tuned on validation removal and rename cases, without labels.
- **Curriculum diff (capability 4).** Courses are matched by name first, then by content (MaxSim $\geq$ the validation threshold for substantial overlap), so a renamed course counts as retained.
- **Topic model (capability 7) and emerging topics (capability 6).** Unit embeddings of training courses are clustered into 40 topics named by class-based TF-IDF; common topics in a report carry these labels. Emerging-topic gaps compare each programme's units with 21 verified topics from cited sources (World Economic Forum Top 10 Emerging Technologies 2025 and 2026, Stanford AI Index 2025 and 2026, AICTE emerging areas); 5 of 67 topic-programme checks are flagged as gaps. The list is a draft pending the group's approval; unverified entries are not used.

2. Results on the test split

Method	Spearman [95% CI]	ROC-AUC [95% CI]	Avg. precision	P / R / F1 at val threshold
TF-IDF cosine (Week 2 baseline)	0.708 [0.62, 0.78]	0.959 [0.93, 0.98]	0.912	0.800 / 0.750 / 0.774
Chunk MaxSim, soft (core)	0.667 [0.55, 0.76]	0.932 [0.88, 0.97]	0.864	0.815 / 0.688 / 0.746
Chunk MaxSim, unit coverage F	0.705 [0.56, 0.83]	0.794 [0.70, 0.89]	0.698	0.944 / 0.531 / 0.680

- **Overlap.** MaxSim reaches Spearman 0.667 against 0.708 for TF-IDF, and ROC-AUC 0.932 against 0.959: no gain for ranking whole courses. NMIMS syllabi re-use wording across programmes, so lexical similarity is already strong; what MaxSim adds is the unit-level explanation (which topics are common and which are unique). Versions of one course (separate task, 359 pairs) score a median 1.000, above substantially overlapping different courses (0.702).
- **Redundancy.** Injected near-duplicate units are ranked in a programme's top 20 in 100% of cases (course-level TF-IDF: 60%). On the test courses 114 unit pairs are flagged.
- **Missing prerequisites.** Removal recall 0.880 (baseline 1.000), false alarms on intact programmes 0.036 (baseline 0.470) and after renaming the prerequisite course 0.120 (baseline 1.000). Flags on the test courses: 33 (baseline 136).
- **Curriculum diff.** With renamed courses, precision rises from 0.793 (names only) to 0.993; on the real year pairs it is 0.998.
- **Flag precision (LLM-annotated reference labels).** Top 20 redundancy flags: precision 0.250 for the course-level baseline and 0.800 for unit-level MaxSim. Missing-prerequisite flags (a sample of 20 per method): precision 0.050 (baseline) and 0.100 (core); most flagged prerequisites are in fact taught earlier under another name or are not subjects at all (analysed in Week 5). Emerging topics: the 5 flagged gaps are all judged correct (precision 1.000), but only 11% of the 46 gaps seen by the labels are flagged, so the threshold is too lenient. Topic labels accepted: 70%. Detector flags (Week 3 onwards) were labelled by two Claude passes with different wording plus a pass 3, with no Phi-3, so agreement on flags is not independent. Claude-vs-Claude agreement on flags: 97% (same model family, not independent). Pair labels: LLM-vs-LLM agreement, reported separately: Claude vs Phi-3 on 106 pairs 74% (weighted kappa 0.561); Claude vs Claude (wording B) on 94 pairs 99% (weighted kappa 0.969). No human annotators were used; results are indicative. For 94 of 200 pairs the second labeller was a Claude prompt rather than Phi-3, so agreement on those items is not independent.

3. Example output (real NMIMS courses, test split)

```
{
  "course_a": {
    "id": "nm48af830b",
    "course_name": "Department Elective IV - Supply Chain Analytics",
    "code": "BTME08016",
```

```

"academic_year": "2021-22",
"semester": "VIII",
"programmes": [
  "MECHANICAL"
]
},
"course_b": {
  "id": "nm9d8c41ec",
  "course_name": "Open Elective I - Logistics and Supply Chain Management",
  "code": "MBOEMEL001",
  "academic_year": "2021-22",
  "semester": "VII",
  "programmes": [
    "MECHANICAL"
  ]
},
"overlap": {
  "percent": "91%",
  "unit_coverage_f": 0.909,
  "semantic_similarity": 0.791,
  "lexical_similarity": 0.698,
  "redundancy_warning": true
},
"common_topics": [
  {
    "topic_a": "Designing the Supply Chain Network",
    "topic_b": "Designing the Supply Chain Network",
    "similarity": 1.0,
    "topic_label": "marketing, supply chain, customer"
  },
  {
    "topic_a": "Warehouse and Transportation Management",
    "topic_b": "Warehouse and Transportation Management",
    "similarity": 0.961,
    "topic_label": "waste, industrial, quality"
  },
  {
    "topic_a": "Building a Strategic Frame Work to Analyze Supply Chain",
    "topic_b": "Building a Strategic Framework to Analyze Supply Chain",
    "similarity": 0.943,
    "topic_label": "marketing, supply chain, customer"
  },
  {
    "topic_a": "Supply chain Analytics-II",
    "topic_b": "IT in Supply Chain",
    "similarity": 0.692,
    "topic_label": "cost, demand, costs"
  },
  {
    "topic_a": "Modelling in Supply Chain Management",
    "topic_b": "Building a Strategic Framework to Analyze Supply Chain",
    "similarity": 0.603,
    "topic_label": "marketing, supply chain, customer"
  }
],
"unique_to_a": [
  "Supply chain Analytics-I"
],
"unique_to_b": []
}

```

4. Findings and limitations

- Unit-level semantics help where the committee needs help: one repeated unit inside otherwise different courses (redundancy), renamed courses (diff, prerequisite false alarms). They do not beat TF-IDF at ranking whole courses, and missing-prerequisite flags are still mostly wrong, which Week 5 analyses.
- Prerequisite checks depend on the programme structure built from the NMIMS semester syllabus booklets, which has gaps where booklets are scanned (531 of 2,413 non-exam PDFs parsed); programmes with little data are skipped. OCR is future work.
- Labels are LLM-annotated reference labels; the label-free tests (injection, removal, renaming) do not depend on them.
- Week 4: encoder and aggregation ablations, score fusion and calibrated confidence, Bloom classifier, with paired confidence intervals.

AI assistance

Code and report text were written with AI assistance (Claude). Reference labels are LLM-annotated (Claude, Phi-3). Every number is generated from `results/*.json` by `tools/build_report.py` and verified by `tools/check_reports.py`.